



G.K. Gujar Memorial Charitable Trust's
Dr. Ashok Gujar Technical Institute's,
Dr. Daulatrao Aher College of Engineering, Karad.
Vidyanagar Ext. Banawadi, Tal. Karad 415124, Dist. Satara, Maharashtra INDIA

Program : Artificial Intelligence & Data Science
Semester: IV

DACOE/ PRJ -FRM-07

PRJ -FRM-07 Rev. No: 0 Date:

Synopsis

1. Title of Project : Real-Time Helmet Detection System Using Computer Vision and ESP32

2. Name of college : Dr. Daulatrao Aher College of Engineering, Karad

3. Name of Department : Artificial Intelligence & Data Science

4.Name of students :
Raj N. Kumbhar
Sanchita G. Kumbhar
Sayali R. Kumbhar
Niranjan N. Mulik

5.Name of guide : Prof S. Y. Inamdar

6.Relevance :

Road safety is a major concern in India, especially for two-wheeler riders. A large number of accidents result in serious head injuries because many riders do not wear helmets. Although helmet use is mandatory by law, enforcement is difficult due to heavy traffic and limited manual monitoring by traffic police. Continuous checking at every signal is not possible, which leads to rule violations and increased risk of fatalities.

This project is relevant because it provides an automated helmet detection system that works in real time without human intervention. By using computer vision and ESP32, the system can immediately identify whether a rider is wearing a helmet and control traffic signals accordingly. This helps in improving rule enforcement, reducing manual effort, and promoting helmet awareness among riders

7. Literature Review :

1.Helmet Detection Using Deep Learning (YOLO/CNN) :

- Many researchers have used deep learning models such as YOLO (You Only Look Once) and CNN (Convolutional Neural Networks) for helmet detection.
- These models provide high accuracy and better detection performance.
- However, they require powerful GPUs, high memory, and expensive hardware.
- They are complex and difficult to implement in low-cost, real-time systems.

2.CCTV-Based Traffic Monitoring Systems

- Some systems use CCTV cameras along with human monitoring to check helmet usage.
- This method is time-consuming and requires continuous human supervision.
- It is not efficient for 24/7 monitoring.
- There are higher chances of human error and delayed response.



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3. Haar Cascade for Face Detection :

- Haar Cascade is a fast and lightweight algorithm used for real-time face detection in OpenCV.
- It works efficiently on low-power systems and does not require heavy computation.
- It is suitable for basic and low-cost helmet detection applications.
- However, its accuracy may decrease in complex real-world conditions.

4. Color-Based Helmet Detection (HSV Model) :

- Some systems detect helmets using color detection in HSV (Hue, Saturation, Value) color space.
- This method is simple, fast, and easy to implement.
- It works well when the helmet color is clearly distinguishable.
- However, its accuracy depends on lighting conditions and helmet color variations.

5. IoT-Based Traffic Signal Control Using ESP32 :

- ESP32 is widely used in IoT-based projects for hardware control and communication.
- It can receive data from a computer system and control traffic signals or LEDs in real time.
- It supports wireless communication and is cost-effective.
- It is suitable for smart traffic management systems.

6. Limitations of Existing Systems :

- Most existing systems are expensive and require high processing power.
- Deep learning-based systems, though accurate, are not suitable for low-budget applications.
- Manual monitoring systems are inefficient and require continuous human effort.
- Some methods lack accuracy under different environmental conditions.
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8. Problem identification :

with strict traffic rules many riders do not wear helmets. Checking manually if people are wearing helmets requires human effort is not efficient at busy intersections and cannot be done 24/7. The challenge here is to design a system that automatically detects if riders are wearing helmets in time is low-cost and can be integrated with hardware to control traffic signals.

9. System Architecture :

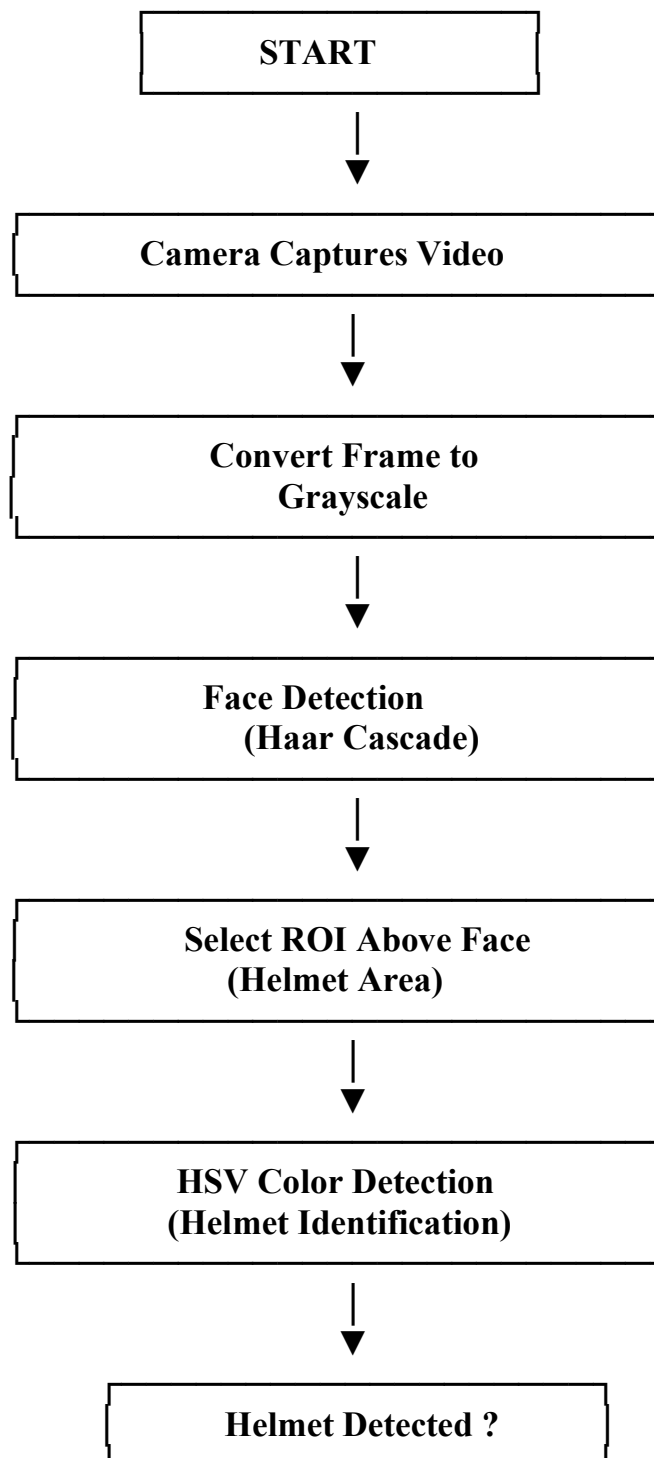


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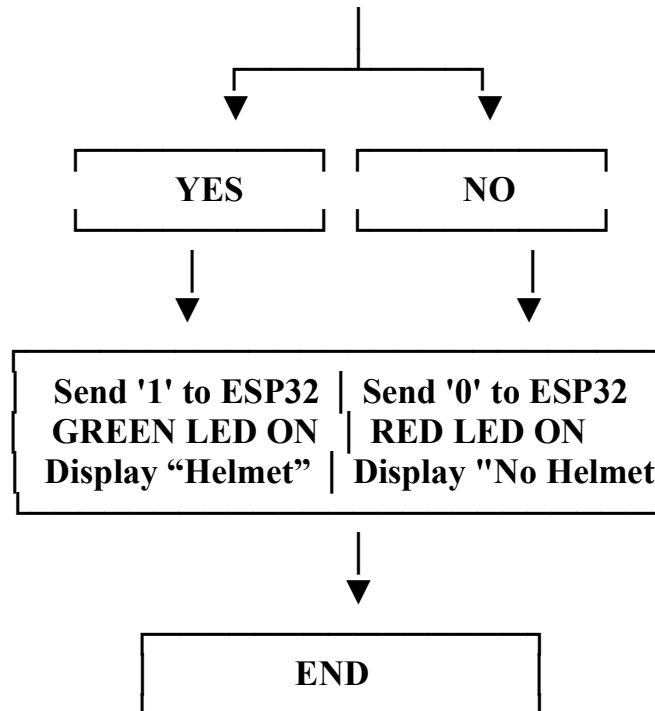


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10.Experimental Setup :

- 1.Hardware: ESP32, LEDs, Webcam, Laptop.
- 2.Software: Python, OpenCV, NumPy, Arduino IDE.

11.Objective :

- To design a real-time helmet detection system using computer vision.
- To detect rider face using Haar Cascade algorithm.
- To identify helmet using HSV color detection.
- To integrate ESP32 for traffic signal control.
- To develop a low-cost road safety solution.

12. Scope of Project :

- Implementation at traffic signals and toll plazas.
- Use in college campus entry monitoring.
- Real-time webcam-based detection.



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- Focus on low-cost hardware.
- Accuracy depends on lighting and helmet color.
- Future upgrade using YOLO/CNN.

13. Proposed work :

The proposed system focuses on developing a real-time helmet detection model using computer vision and ESP32 hardware. The system will use a webcam to capture live video frames, which will be processed using OpenCV in Python. Face detection will be performed using the Haar Cascade algorithm, and the region above the detected face will be selected as the area of interest for helmet detection. This region will be converted into HSV color space to identify helmet colors based on predefined thresholds. After processing, the system will determine whether a helmet is present or not and classify the result accordingly.

The detection result will be transmitted to the ESP32 through serial communication. Based on the received signal, the ESP32 will control RED and GREEN LEDs to indicate helmet and no-helmet conditions. The system will also display the detection output on the screen for monitoring purposes. The hardware setup will be designed to be low-cost and easy to install. The system will be tested under different lighting conditions, distances, and helmet colors to evaluate performance. The main goal is to create a simple, real-time, and efficient solution that can be used at traffic signals, college gates, and small monitoring systems.

14. Motivation for work :

In India, many two-wheeler riders do not wear helmets even though it is compulsory by law. A large number of road accident deaths in India happen due to head injuries, and many of these cases involve two-wheeler riders who were not wearing helmets. Lack of helmet usage, negligence, and absence of continuous monitoring at traffic signals are major causes of these deaths. Manual checking by traffic police is not possible

at all locations and at all times, especially during heavy traffic. This project is motivated by the idea of improving road safety using affordable technology. By combining computer vision with ESP32 hardware, the system can automatically detect whether a rider is wearing a helmet and control



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traffic signals accordingly. This low-cost solution can help traffic authorities, reduce rule violations, and create awareness among riders to wear helmets

14. Expected Outcome :

- It will show Helmet / No Helmet on screen.
- GREEN light will turn ON when helmet is detected.
- RED light will turn ON when helmet is not detected.
- Python will send data to ESP32 successfully.
- The system will work automatically without human help.
- It will save time of traffic police.
- The setup will be low cost and simple.
- It will work properly in normal light conditions.
- It can run for long time continuously.
- It can be used at traffic signals and college gates.
- It can be improved in future using AI models.

15. Expected Date of Completion :

16. References :

1. OpenCV Official Documentation, Available: <https://opencv.org>
2. Python Software Foundation, Python Documentation, Available: <https://www.python.org>
3. Espressif Systems, ESP32 Technical Reference Manual, Available: <https://www.espressif.com>
4. Arduino IDE Documentation, Available: <https://www.arduino.cc>
5. Viola, P. and Jones, M., "Rapid Object Detection using a Boosted Cascade of Simple Features," IEEE, 2001.
6. World Health Organization (WHO), "Global Status Report on Road Safety," 2018.
7. Ministry of Road Transport and Highways (MoRTH), Government of India, "Road Accidents in India Report."
8. Research papers on Helmet Detection using Computer Vision and Deep Learning.



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Roll No.	Name of Student	Signature
240105021	Raj Nitin Kumbhar	
240105022	Sayli Ratan Kumbhar	
240105023	Sanchita Ganesh Kumbhar	
240105031	Niranjan Nitin Mulik	